

Phillip W. Yoon

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EDUCATION

Hunter College, City University of New York <i>M.A. in Mathematics.</i>	2025–2027 (expected) New York, NY
University of Pennsylvania, Graduate School of Education <i>M.S.Ed. in Independent School Teaching Residency</i>	2022–2024 Philadelphia, PA
Northern Illinois University <i>Graduate (M.A.) coursework in philosophy, 18 credits (non-degree)</i>	2021–2022 DeKalb, IL
Johns Hopkins University <i>B.A. in Mathematics and Philosophy, General Honors.</i> <i>Study abroad: Budapest Semesters in Mathematics, Summer 2019.</i>	2017–2021 Baltimore, MD

RESEARCH INTERESTS

Algebra and combinatorics, with current work in commutative and homological algebra, algebraic combinatorics, and non-commutative algebra. Additional interests include topological methods in algebra, representation theory, and solvable lattice models.

PUBLICATIONS

1. Noah Ripke and **Phillip Yoon**, “Characteristic Independence of Betti Numbers of Monomial Ideals in Five Variables,” *Journal of Pure and Applied Algebra* **230** (2026), 108354. arXiv:2607.10639.

RESEARCH EXPERIENCE

NSF REU/RET in Mathematics, California State University, Chico <i>Researcher, Commutative Algebra and Combinatorics Group</i>	June–August 2026 Chico, CA
• Conducted research under the supervision of Guillermo Alesandroni on characteristic dependence of Betti numbers of monomial ideals. • Proved with a coauthor that the multigraded, graded, and total Betti numbers of monomial ideals in five variables are independent of the characteristic of the base field; led the drafting and revision of the resulting manuscript. • Developed a homological and combinatorial proof based on Hochster’s formula; wrote Python code reducing 7,580 squarefree ideals to 210 representatives up to variable relabeling and carried out computational verification in Macaulay2. • Continuing this project by investigating characteristic dependence in seven and eight variables using combinatorial and topological methods.	
Hunter College, City University of New York <i>Independent Research in Combinatorics on Words and Noncommutative Algebra</i>	October 2025–present New York, NY
• Conducting ongoing joint research with Be’eri Greenfeld on growth phenomena in associative and monomial algebras and their connections with combinatorics on words. • Developed results using hereditary languages and combinatorial constructions of languages with prescribed growth properties; manuscript in preparation. • Presented part of this work at Hunter College’s student research colloquium.	
Polymath Jr., University of Minnesota <i>Research Participant</i>	June–August 2025 Remote
• Investigated recursive partition functions for lattice models related to quantum superalgebras under the mentorship of Ben Brubaker. • Contributed to a collaborative project involving color and supercolor variants and local recurrence relations; presented this work at the 2026 Joint Mathematics Meetings.	

RESEARCH TALKS AND PRESENTATIONS

“Characteristic Independence of Betti Numbers of Monomial Ideals in Five Variables” Invited seminar talk , CUNY Commutative Algebra and Algebraic Geometry Seminar, The Graduate Center, CUNY, New York, NY. Joint work with Noah Ripke.	November 2026
“Characteristic Independence of Betti Numbers of Monomial Ideals in Five Variables” Invited talk , Route 81 Conference on Commutative Algebra and Algebraic Geometry, Syracuse University, Syracuse, NY. Joint work with Noah Ripke.	September 2026

“What Does Small Growth Mean for Algebras?” Hunter Mathematics and Statistics Colloquium: Student Research Presentations, Hunter College, New York, NY.	April 2026
“Lattice Models for Quantum Superalgebras: Color and Supercolor” Joint Mathematics Meetings, Washington, DC	January 2026
“Finding Equations for Parametrized Curves of an Affine Variety” Poster presentation , CUNY Directed Reading Program, The Graduate Center, CUNY, New York, NY.	May 2025

ADDITIONAL MATHEMATICAL EXPERIENCE

CUNY Directed Reading Program, The Graduate Center <i>Algebraic Geometry and Computational Commutative Algebra</i>	January–May 2025 New York, NY
Read selected material from Cox, Little, and O’Shea’s <i>Ideals, Varieties, and Algorithms</i> with graduate mentor Valeriy Sergeev; completed an expository project and final presentation.	

TEACHING EXPERIENCE

Hunter College, City University of New York <i>Adjunct Lecturer</i>	August 2026–present New York, NY
Instructor of record for Precalculus (MATH 12550).	
Columbia Grammar and Preparatory School <i>Teacher of Mathematics</i>	August 2025–June 2026 New York, NY
Courses taught: AP Calculus AB and Precalculus.	
New York City College of Technology, CUNY <i>Adjunct Lecturer</i>	August–December 2024 Brooklyn, NY
Courses taught: College Algebra and Trigonometry (MAT 1275CO) and Precalculus (MAT 1375).	
Trinity School <i>Mathematics Teaching Fellow</i>	August 2022–June 2024 New York, NY
Courses taught: Prealgebra and Honors Algebra I.	
Johns Hopkins University <i>Teaching Assistant and Tutor</i>	August 2018–May 2021 Baltimore, MD
Supported Discrete Mathematics (EN.553.171), Probability and Statistics (EN.553.211), Calculus II (AS.110.109), and Linear Algebra (AS.110.201).	

AWARDS AND HONORS

- John Arthur Loustau Scholarship, Hunter College, CUNY, 2026.
- Gail J. McGovern Scholarship, Johns Hopkins University, 2018–2020.
- Honorable Mention, International Olympiad on Astronomy and Astrophysics, 2016.

SERVICE AND OUTREACH

- Member, AI Committee, Columbia Grammar and Preparatory School, 2025–2026.
- MATHCOUNTS Coach, Trinity School, 2022–2024.
- Co-organizer, Johns Hopkins Mathematics Tournament, 2018–2020.

LANGUAGES AND TECHNICAL SKILLS

Languages: English, Korean, Japanese.

Technical: $\text{\LaTeX}$, Python, Macaulay2.

ADDITIONAL PROFESSIONAL EXPERIENCE

xAI <i>AI Tutor</i>	January–August 2025 Remote
Evaluated and improved AI-generated mathematical reasoning and written solutions.	